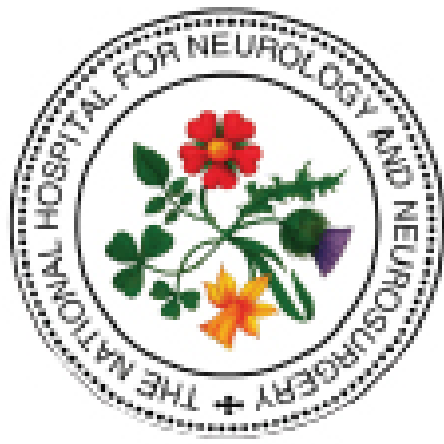




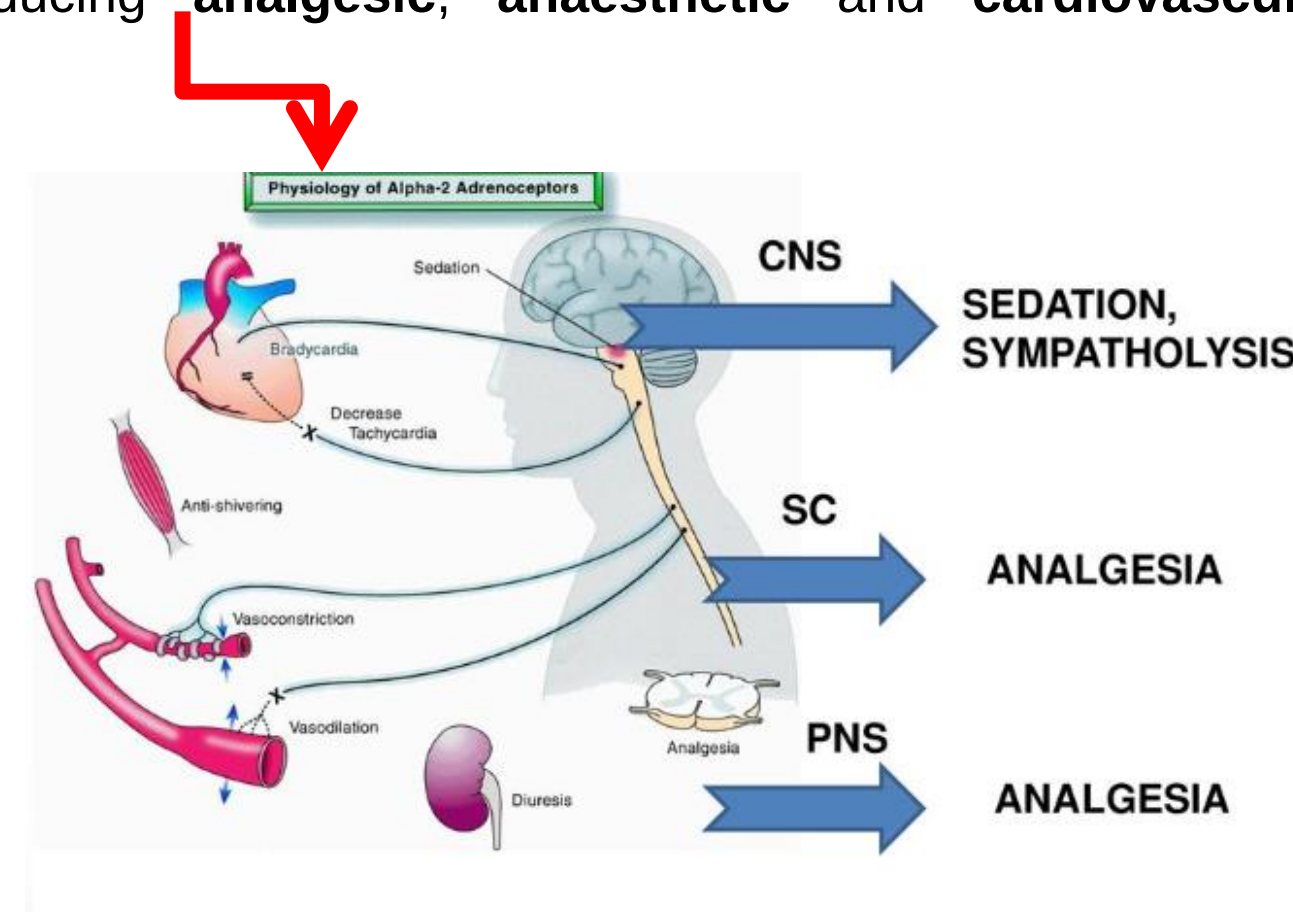
The National Hospital for Neurology and Neurosurgery Neuroscience Critical Care Unit (NCCU)

DEXMEDETOMIDINE

LIGHT SEDATION in MECHANICALLY VENTILATED PATIENTS to FACILITATE WEANING and EXTUBATION

Dexmedetomidine is a selective alpha-2 receptor agonist and produces clinical effects after binding to G-Protein-coupled α_2 -AR, of which there are three subtypes (α_2A , α_2B , and α_2C)

Each has different physiological functions and pharmacological activities. These receptor subtypes are found in the central, peripheral and autonomic nervous systems, as well as in vital organs and blood vessels, producing **analgesic**, **anaesthetic** and **cardiovascular** effects



PHARMACOKINETICS

- Distribution phase rapid, with half-life of distribution approximately 6min and elimination half life 2h
- Onset of effect within 15min / peak effect at 60min
- Metabolised in liver and excreted in urine (95%) and faeces (4%)
- Does not suppress respiratory centre

INDICATIONS FOR USE

NCCU CONSULTANT INITIATION ONLY

- Light sedation (RASS 0 to -3) in mechanically ventilated patients
- For patients with actual or suspected hyperactive delirium where agitation precludes weaning and extubation
- Only used after standard sedative agents (including propofol, clonidine or benzodiazepine) trialled for 48h

ADVERSE EFFECTS / CAUTIONS

- Most frequently reported adverse reactions on commencing are hypotension and bradycardia – *reduce rate by 0.1mcg/kg/h or stop infusion*
- For patients with hepatic impairment starting dose reduced to 0.4mcg/kg/h and may need reduced maintenance dose
- No dose adjustment needed in renal impairment
- Hypersympathetic withdrawal symptoms reported with abrupt discontinuation of prolonged infusions

MONITORING

- Cardiac and respiratory monitoring In all patients
- 1 hourly monitoring of RASS and 4 hourly monitoring of GCS to assess for over-sedation

PREPARATION

Each 10mL vial contains 1000mcg dexmedetomidine

Take 2mL (200mcg) from vial and dilute with 0.9% sodium chloride or glucose 5% to form solution of 50mL with a concentration of **4mcg/mL** (chemically / physically stable for 24h at 25°C but should be used immediately (microbiology perspective))

STEP 1

INFUSION

When plan made to extubate patient start dexmedetomidine infusion and plan to wean other sedative drugs - No loading dose required – **DO NOT BOLUS**

- Initial infusion rate **0.7mcg/kg/h** (based on **ideal** body weight) - titrated to RASS and adjusted **up or down** according to response (usual dose 0.2–1.4mcg/kg/h)
- Dose can be titrated up to **maximum 1.4mcg/kg/h** to achieve desired level of sedation
- After dose adjustment, a new steady state sedation level may not be reached for up to **1 hour**
- If significant hypotension / bradycardia on starting infusion – *reduce rate by 0.1mcg/kg/h or stop*
- In **hepatic impairment** starting dose reduced to **0.4mcg/kg/h** +/- reduced maintenance dose

STEP 2

WEANING OTHER SEDATIVES / OPIOIDS

Attempt weaning after **2h** - Titrate to RASS

- *Initially halve other drugs – e.g. propofol and fentanyl*
- *Then halve these drugs again within ~ 2h*
- *Then stop these drugs after further ~ 2h*
- If patient becomes agitated increase dexmedetomidine dose by 0.1mcg/kg/h
- Consider addition of olanzapine 2.5 - 5mg 6 hourly
- Haloperidol not recommended due to increased risk of delirium

STEP 3

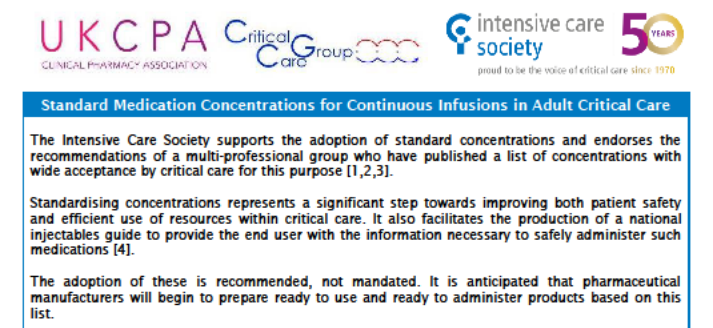
WEAN DEXMEDETOMIDINE as able once extubated – titrate to RASS

Reduce by half initially
Reduce by half again in ~ 2h
Then stop after further ~ 2h

DURATION OF INFUSION SHOULD NOT EXCEED 72 HOURS

DESIRED DOSE mcg/kg/h	MAINTENANCE INFUSION RATE (mL/h) of 4mcg/mL SOLUTION											
	PATIENT WEIGHT (kg) (Ideal Body Weight – IBW)											
	40	45	50	55	60	65	70	75	80	85	90	100
0.2	2	2.3	2.5	2.8	3	3.3	3.5	3.8	4	4.3	4.5	5
0.4	4	4.5	5	5.5	6	6.5	7	7.5	8	8.5	9	10
0.6	6	6.8	7.5	8.3	9	9.5	10.5	11.3	12	12.8	13.5	15
0.7	7	7.9	8.8	9.7	10.5	11.4	12.3	13.1	14	14.9	15.8	17.5
0.8	8	9.0	10	11	12	13	14	15	16	17	18	20
1.0	10	11.3	12.5	13.8	15	16.3	17.5	18.8	20	21.3	22.5	25
1.2	12	13.5	15	16.5	18	19.5	21	22.5	24	25.5	27	30
1.4	14	15.8	17.5	19.3	21	22	24.5	26.3	28	29.8	31.5	35

REFERENCES



Medication	Concentration	Example Infusion Concentration	Central or Peripheral
Morphine	1mg/mL	50mg in 50mL	C / P
	2mg/mL	100mg in 50mL	C / P
Fentanyl	50micrograms/mL	2.5mg in 50mL	C / P
Alfentanil	500micrograms/mL	25mg in 50mL	C / P
Remifentanyl	50micrograms/mL	2mg in 40mL	C / P
	100micrograms/mL	5mg in 50mL	C / P
Midazolam	1mg/mL	50mg in 50mL	C / P
	2mg/mL	100mg in 50mL	C / P
Clonidine	15micrograms/mL	750micrograms in 50mL	C / P
	4micrograms/mL	200micrograms in 50mL	C / P
Dexmedetomidine	8micrograms/mL	400micrograms in 50mL	C / P
	10micrograms/mL	500micrograms in 50mL	C / P
Adrenaline	10micrograms/mL	5mg in 50mL	C / P
	320micrograms/mL	16mg in 50mL	C / P
	16micrograms/mL	32mg in 100mL	C / P
	16micrograms/mL	4mg in 250mL	C / P
Noradrenaline	80micrograms/mL	4mg in 50mL	C
	16micrograms/mL	8mg in 50mL	C
	16micrograms/mL	16mg in 100mL	C
	320micrograms/mL	16mg in 50mL	C
	320micrograms/mL	32mg in 100mL	C
Vasopressin (Arginine)	0.4units/mL	20units in 50mL	C / P
Dobutamine	5mg/mL	250mg in 50mL	C

Version 4.1

December 2020

<https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://ics.ac.uk/resource/standard-concentrations.html&ved=2ahUKEwiDtavskNGHAXaUaV0EAHQqoFVwQFnoECBUQAQ&usg=AOvVaw2SUQ2osmCcnMDICCVypbC5>

1. The effect of dexmedetomidine on delirium and agitation in patients in intensive care: systematic review and meta-analysis with trial sequential analysis Anaesthesia 2019 Mar;74(3):380-392.
<http://dx.doi.org/10.21037/atm.2016.05.44>
2. Delirium in the Intensive Care Unit: Is Dexmedetomidine Effective? Joelle Ungarian, James A. Rankin, Karen L. Then. *Critical Care Nurse*. 2019;39[4]

INITIAL INFUSION RATE
Based on ideal body weight